

CLIMATE DATA ANALYSIS · 2026

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Covering 1750 – 2024 · Global Carbon Budget 2025 · NASA / NOAA Mauna Loa

429 ppm

Atmospheric CO₂ · Feb 2026

35+ Bt

Annual fossil fuel CO₂ · 2024

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

ppm CO₂ · Feb 2026

Mauna Loa Observatory (NOAA)

~280 ppm

Pre-industrial baseline (1750)

+53%

Increase since industrialization

20,000 yrs

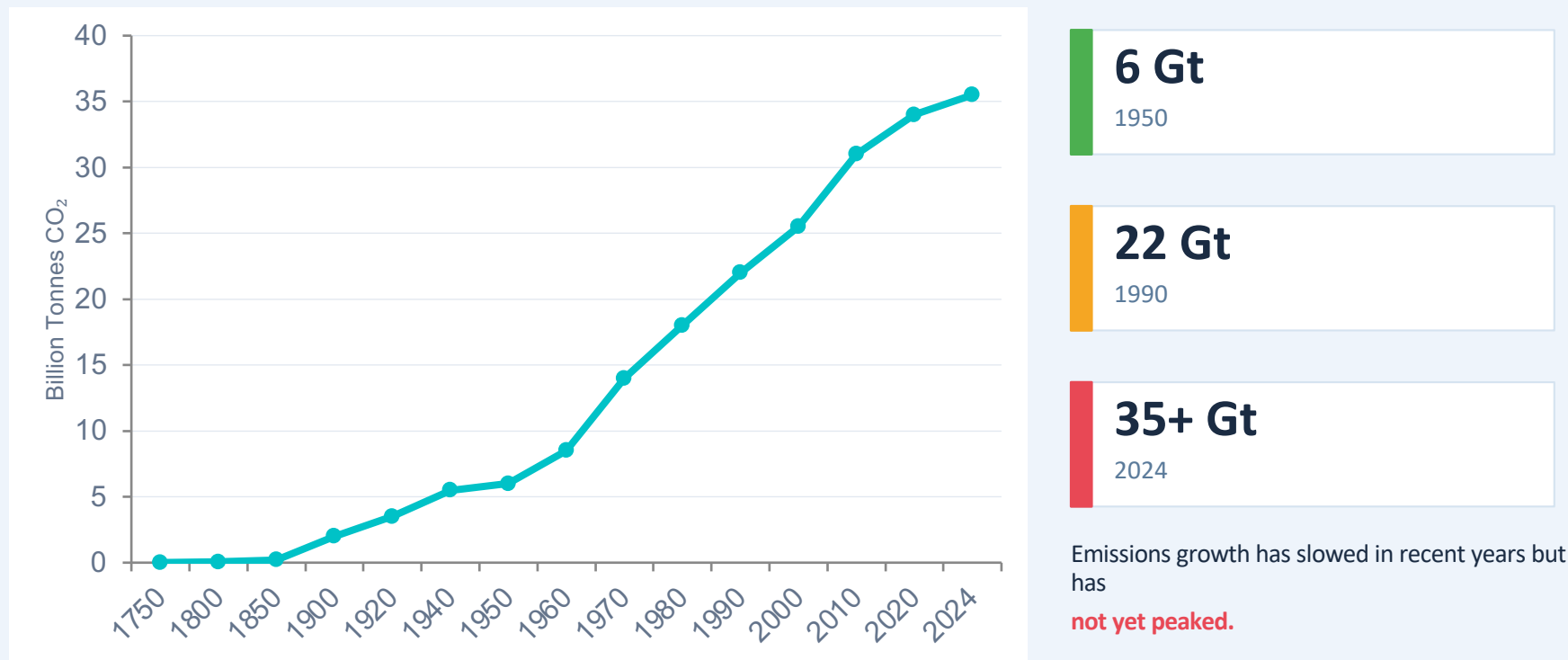
Natural ice-age change took this long

~200 yrs

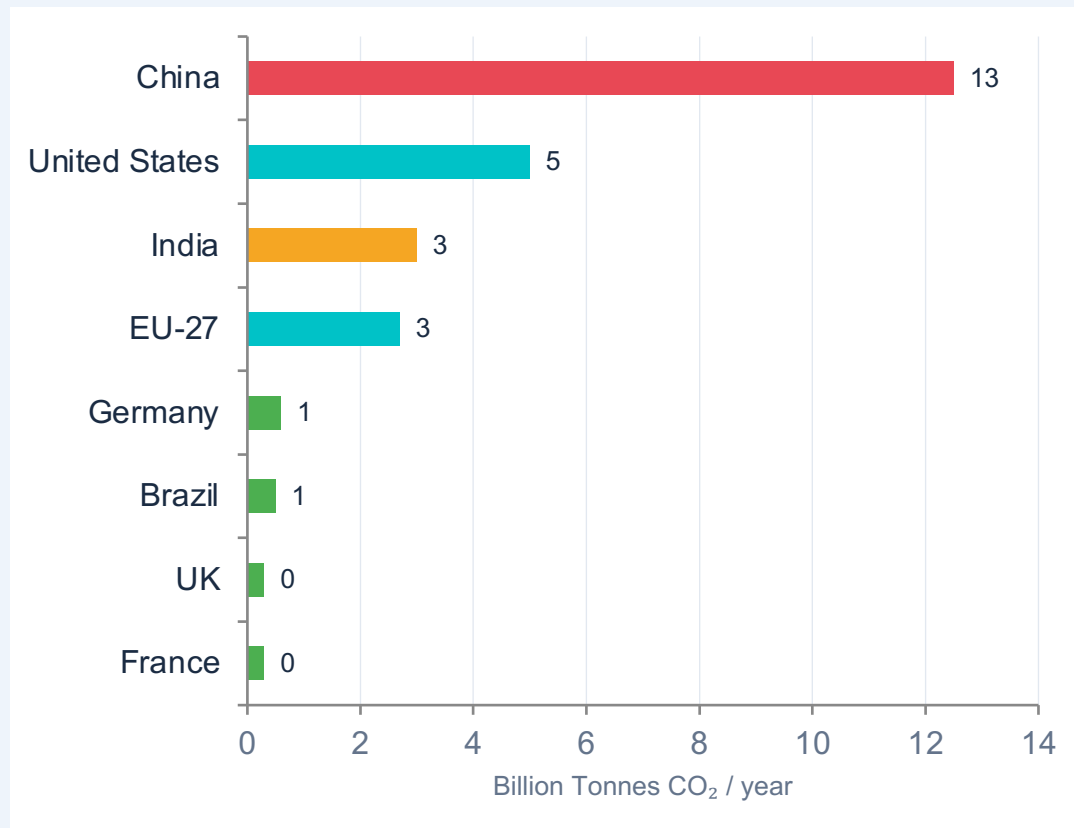
Human-driven change has taken only this

Current CO₂ levels are at least **1.5× higher** than the natural increase observed after the last ice age — a change that previously took **thousands of years** but has occurred in roughly **two centuries** of industrialization. Ground-based and satellite measurements confirm the sharp acceleration, linked primarily to the burning of coal, oil, and natural gas.

Global Fossil Fuel CO₂ Grew Six-Fold: 6 Billion to 35+ Billion Tonnes (1950–2024)



China Now Emits Over 12 Billion Tonnes — More Than Double the United States



China — ~12.5 Bt

~27% of global total · slight increase

United States — ~5.0 Bt

~14% · declining from ~6 Bt peak (2005)

India — ~3.0 Bt

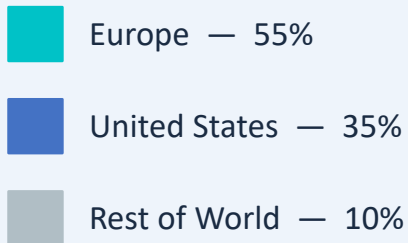
~8% · rapidly rising

EU-27 — ~2.7 Bt

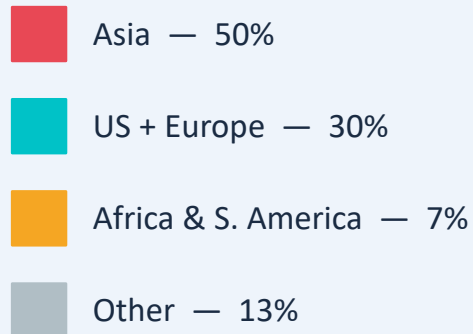
~7% · declining

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900



2024



Shift in 120 years: Europe + US went from >90% to <33% of global emissions. Asia rose from <10% to ~50%. Africa and South America each contribute only 3–4%.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

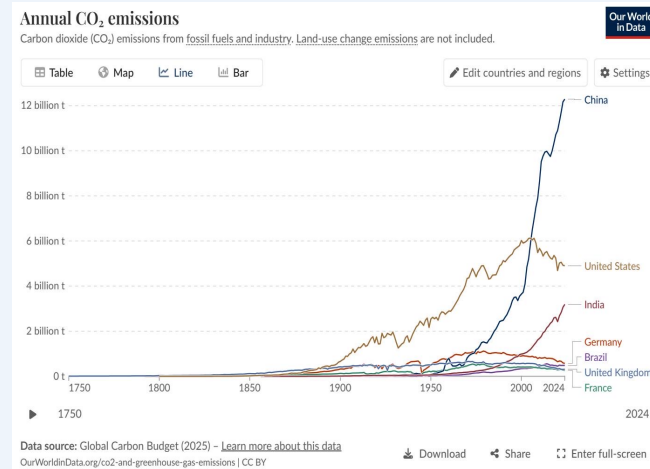
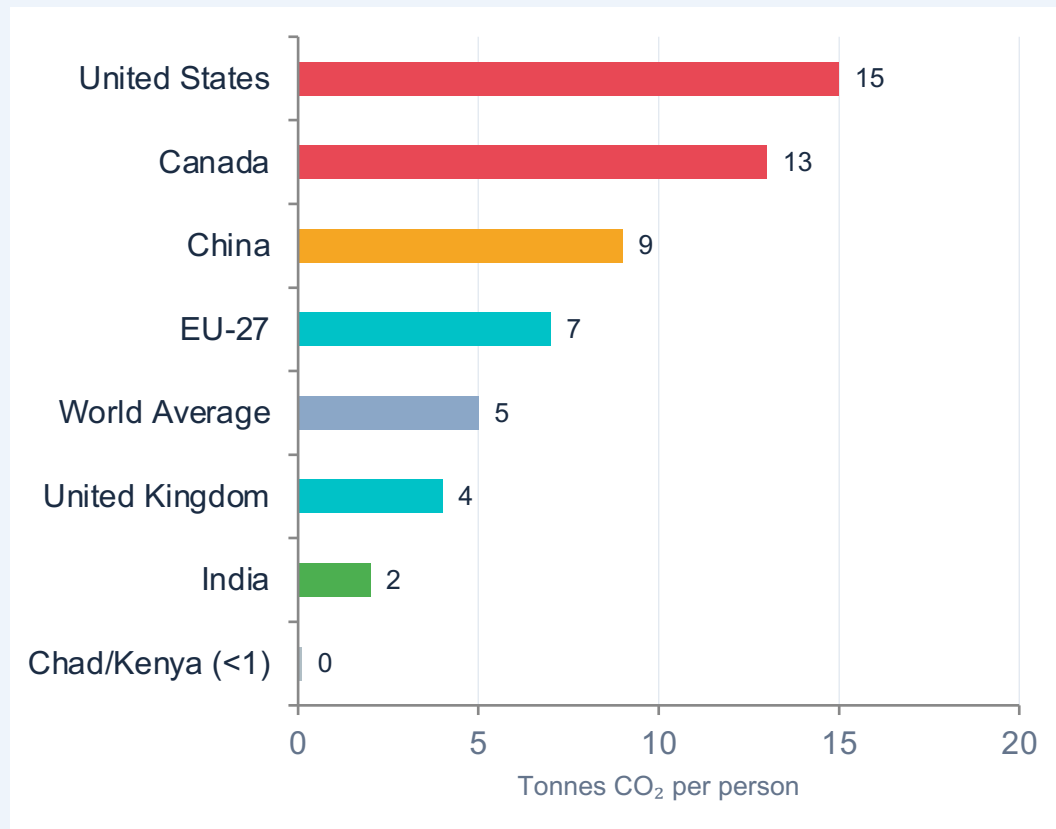


Fig. 3 — Annual CO₂ by country (Our World in Data)

150×

disparity between
highest and lowest
per capita emitters

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual percentage change in CO₂ emissions, 2024

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

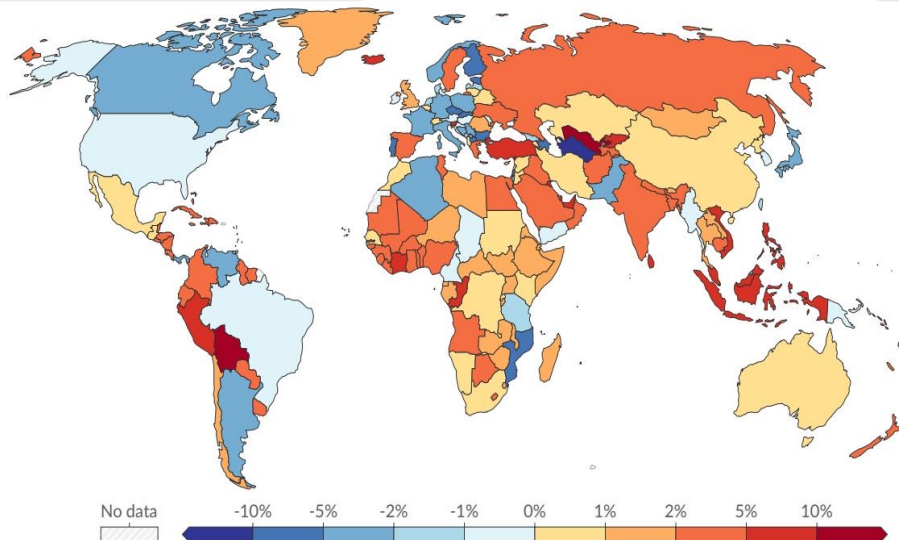
Table

Map

Line

Bar

Select countries



1751

2024

Data source: Global Carbon Budget (2025) - [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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2024 Regional Trends

United States ▼ -1% to -5%

Most of Europe ▼ -1% to -5%

Russia ▲ +2% to +10%

SE Asia ▲ +2% to +10%

Parts of Africa ▲ +2% to +10%

S. America (select) ▲ >+10%

Blue = decline · Orange/Red = increase

Fig. 1 — Annual % change in CO₂ emissions, 2024 (Our World in Data)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita CO₂ emissions — the key variable is their energy source mix.

Country	Per Capita CO ₂	Nuclear Share	Renewables Share	Fossil Fuel Reliance
 France	~4 t/person	High (~70%)	Growing	Low
 United Kingdom	~4 t/person	Moderate	High (wind)	Declining
 Germany	~7–8 t/person*	Low (phase-out)	Growing	Still high
 Netherlands	~8 t/person*	None	Growing	High (gas-heavy)

Green shading = lower per capita CO₂ / low fossil reliance

Amber shading = higher per capita CO₂ / fossil-heavy mix

Key insight: France and Germany have comparable living standards, yet France emits roughly **half as much CO₂ per person** — a direct result of energy policy choices, not prosperity. Policy and energy mix, not wealth alone, determine per capita outcomes.

* Germany ~0.6 Bt total / ~72M pop ≈ ~8t; Netherlands ~0.15 Bt / ~18M pop ≈ ~8t · Source: Global Carbon Budget 2025

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01 Global emissions exceed 35 Bt — not yet peaked

Fossil fuel CO₂ now surpasses 35 billion tonnes/year. Growth has slowed but annual emissions continue to rise.

02 China alone emits >25% of the global total

At ~12.5 Bt, China is more than double the US (~5 Bt). India (~3 Bt) is the third largest and rising fast.

03 Geographic center of emissions has shifted east

Europe + US: from >90% in 1900 to <33% today. Asia now accounts for ~50% of global emissions.

04 Per capita inequalities remain extreme: 150× gap

The US and Australia (~15 t/person) vs. Chad or Niger (~0.1 t/person). Those who emit least face the greatest climate risks.

05 Energy mix — not prosperity — drives per capita outcomes

France vs. Germany illustrates how policy choices on nuclear and renewables create vastly different emission profiles at comparable living standards.

Data Sources

Global Carbon Budget 2025

NASA / NOAA Mauna Loa Observatory

Our World in Data — CO₂ & GHG Emissions